MARKET MAKERS AS ECOSYSTEM ENGINEERS: AN ODE ANALYSIS OF SURVIVAL AND COLLAPSE

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Abstract

Market makers are firms that are always willing to buy and sell in financial markets. They earn the bid–ask spread, but by competing with each other they make that spread smaller. A smaller spread attracts more trading, but leads to less profit per trade. Like ecosystem engineers in biology, market makers thus change their own environment. We ask when the market-making industry can sustain itself, and whether it can suddenly collapse. We model trading interest and market-making presence with a system of ODEs, in which the spread tightens as the market-making presence grows. After nondimensionalization, we determine the equilibria and their stability and study the bifurcations numerically. A dimensionless number R, which we call the survival number, decides whether a first market maker in an empty market grows or shrinks. For $R > 1$, a liquid market is always stable. When trading volume responds strongly to tighter spreads, both an illiquid and a liquid market can be stable for $R < 1$. The industry then survives only if it is already large enough, and a small change in conditions can make it collapse abruptly while the spread is still tight. After collapse, restoring the conditions does not restore the industry.

1 Introduction

A market maker is a firm that is always willing to buy and sell in financial markets. It quotes both a buy and a sell price, and it earns the gap between them. That gap is the bid–ask spread. At the same time, it is the cost that everyone else pays in order to trade. By doing this, market makers play an important role as they provide liquidity to financial markets. In US stock markets, for example, market makers execute between 83 and 99.9 per cent of the shares in retail orders, depending on the order type (Anand et al., 2026). Retail traders therefore rarely trade with each other, but almost always with a market maker.

The bid–ask spread is not a fixed property of the market, but the market makers set it themselves. This makes market makers look a lot like ecosystem engineers in biology. An ecosystem engineer is an organism that changes its environment in a way that changes the availability of resources (Jones et al., 1994). For instance, a mussel bed slows the water down, so more food settles where the mussels can reach it. Market makers do something similar. Competition between market makers pushes the bid–ask spread down, which lowers the cost of trading for everyone else and subsequently raises trading volume.

While smaller spreads bring in more trading volume, they also lower the profit per trade for the market makers. The market makers therefore make their own business harder as the industry grows. Which of the two effects wins decides what happens to the industry, and that is what we model. We ask two questions: *when does a market-making industry sustain itself, and can it collapse abruptly?* The first question matters for new or small markets, which only become liquid if market makers find it worthwhile to enter and stay. The second matters because a gradual decline gives traders and regulators time to react, while a sudden collapse does not. Markets have lost their liquidity suddenly before: during the flash crash of 6 May 2010, a significant number of market makers withdrew from the US equity markets completely within minutes (CFTC and SEC, 2010).

We build a model with two variables: trading interest and market-making presence, by which we mean the size of the market-making industry. The spread is the part of the environment that the market makers change. Section 2 derives the model and writes it in dimensionless form. Section 3 determines the equilibria and their stability, and shows with bifurcation diagrams and phase portraits how they change with the parameters. Section 4 translates these results back to the market and answers the two questions, and Section 5 summarizes our findings and discusses the limitations of the model and directions for further research.

2 A model of market makers as ecosystem engineers

2.1 Model derivation

To describe market makers as ecosystem engineers, we first need a resource and an engineer. The resource is the trading interest I: the orders that traders want to execute but that have not been executed yet. The engineer is the market-making presence M, which consumes the trading interest by trading with it. The environment that the engineer changes is the bid–ask spread s. We assume a homogeneous market in which every trade involves one unit of trading interest and one market maker, and we ignore entry costs and regulation. The model is built from four processes:

$\varnothing \xrightarrow{I_0} I$	New trading interest arrives at a constant rate
$I \xrightarrow{\mu} \varnothing$	Unused trading interest expires
$I + M \xrightarrow{\beta} (1+\gamma s)M$	A trade uses up one unit of interest. The market maker earns the spread and turns it into γs new units of presence
$M \xrightarrow{\delta} \varnothing$	Market makers leave the market at a constant rate

Adding up the four processes gives

$$\begin{cases} \dot{I} = I_0 - \beta I M - \mu I \\ \dot{M} = \gamma s\, \beta I M - \delta M \end{cases} \tag{1}$$

The ecosystem engineering shows through s and β, which both depend on the market-making presence. The spread becomes smaller as more market makers compete, and we take

$$s(M) = \frac{s_0}{1+\alpha M}, \tag{2}$$

so that $s(0) = s_0$ is the spread in a market without market makers and α measures how fast the spread tightens as the presence grows. Here γ is the efficiency with which profit is turned into a larger market-making presence. A smaller spread makes trading cheaper, so trading volume increases. We model this with

$$\beta(s) = \beta_0 \left(1 + c\left(\frac{s_0}{s} - 1\right)\right) = \beta_0(1 + c\alpha M), \tag{3}$$

where β_0 is the trading rate at the baseline spread s_0. The parameter c, which we call liquidity response, measures how strongly the trading rate reacts to a tighter spread. For $c = 0$ the spread has no effect on the trading rate, and for $c = 1$ the trading rate is inversely proportional to the spread. Substituting s and β into (1) gives

$$\begin{cases} \dot{I} = I_0 - \beta_0(1 + c\alpha M) I M - \mu I \\ \dot{M} = \gamma s_0 \beta_0 \dfrac{1 + c\alpha M}{1 + \alpha M} I M - \delta M \end{cases} \tag{4}$$

In system (4), the growth of M is the product of the number of trades, the profit per trade and the efficiency γ. A larger presence makes the spread smaller, which raises the trading rate but lowers the profit per trade. The net effect is described by the fraction in (4), which we denote by

$$h(x) = \frac{1 + cx}{1 + x}, \qquad x = \alpha M. \tag{5}$$

For $c < 1$ the lost margin outweighs the extra volume, so a larger industry earns less per market maker and grows more slowly. For $c > 1$ the extra volume outweighs the lost margin, so a larger industry earns more per market maker and grows faster. Table 1 gives an overview of the parameters.

Table 1: The eight parameters of the dimensional model (4) and their meaning.

Symbol	Meaning
I_0	Inflow of new trading interest per unit of time.
μ	Rate at which unused trading interest expires.
β_0	Baseline trading rate: trades per unit of trading interest and per unit of presence, per unit of time, at spread s_0.
c	Liquidity response: how strongly the trading rate reacts to a tighter spread.
s_0	Spread in a market without market makers.
α	Sensitivity of the spread to market-making presence.
γ	Efficiency with which profit is turned into a larger presence.
δ	Rate at which market makers leave the market.

2.2 Nondimensionalization

To analyse the system, we write it in dimensionless form, so that the results do not depend on the currency or time unit used. We set $I = \kappa_I \tilde{I}$, $M = \kappa_M \tilde{M}$ and $t = \kappa_t \tilde{t}$, substitute this into (4), and choose

$$\kappa_I = \frac{I_0}{\mu}, \qquad \kappa_M = \frac{1}{\alpha}, \qquad \kappa_t = \frac{1}{\mu}. \tag{6}$$

Without market makers, (4) reduces to $\dot{I} = I_0 - \mu I$, so the trading interest settles at I_0/μ, the level where the inflow of new interest equals the amount that expires. With $\kappa_I = I_0/\mu$, $\tilde{I}$ is therefore the trading interest as a fraction of this level: $\tilde{I} = 1$ in a market without market makers, and $\tilde{I} < 1$ when market makers trade part of it away. With $\kappa_M = 1/\alpha$ the spread becomes $s/s_0 = 1/(1+\tilde{M})$, so $\tilde{M}$ measures how much the market makers have tightened the spread: at $\tilde{M} = 1$ the spread has halved. With $\kappa_t = 1/\mu$, one unit of $\tilde{t}$ is the average time that unused trading interest stays in the market before it expires. We could also have measured time by the exit rate of market makers ($\kappa_t = 1/\delta$). This would only change the unit of time and not the results, but it would put the large factor $1/\varepsilon$ in front of the $\tilde{I}$ equation, with $\varepsilon = \delta/\mu$ small because unused trading interest usually disappears within hours, while market makers typically stay in a market much longer (Anand et al., 2026). We prefer the small factor ε in front of the $\tilde{M}$ equation, because it is easier to interpret. Throughout this report we assume $\varepsilon \ll 1$. Then $\tilde{M}$ changes much more slowly than $\tilde{I}$, so the system is a fast–slow system. On the timescale of $\tilde{I}$, the market-making presence is almost fixed. Three dimensionless groups remain,

$$b = \frac{\beta_0}{\mu\alpha}, \qquad \varepsilon = \frac{\delta}{\mu}, \qquad R = \frac{\gamma s_0 \beta_0 I_0}{\mu\delta}, \tag{7}$$

and the system becomes

$$\begin{cases} \dfrac{\mathrm{d}\tilde{I}}{\mathrm{d}\tilde{t}} = 1 - b\,(1 + c\tilde{M})\,\tilde{I}\tilde{M} - \tilde{I} \\ \dfrac{\mathrm{d}\tilde{M}}{\mathrm{d}\tilde{t}} = \varepsilon\,\tilde{M}\left(R\,h(\tilde{M})\,\tilde{I} - 1\right) \end{cases} \tag{8}$$

The number R is central to the model. Its numerator $\gamma s_0 \beta_0 I_0/\mu$ is the growth rate of a first market maker in an empty market: it trades with the trading interest I_0/μ that is present, at rate β_0, earns the wide spread s_0 on every trade, and turns that profit into a larger presence with efficiency γ. The denominator δ is the rate at which market makers leave. So $R > 1$ means that a first market maker grows faster than it leaves, and $R < 1$ that it shrinks. This is why we call R the survival number.

The scaling has reduced the eight dimensional parameters of Table 1 to the four dimensionless ones in Table 2. Two of these describe the market itself: R says whether market making pays off at all, and b says how fast the market makers use up trading interest, relative to how fast it expires. The other two describe the liquidity response and the timescales: c sets how strongly the trading rate reacts to a tighter spread, and ε compares how fast market makers leave with how fast trading interest expires. Since ε multiplies the whole $\tilde{M}$ equation, it changes the speed of the dynamics but not the position of the equilibria. Note also that γ and s_0 appear only as the product γs_0, so the model cannot separate a wide spread with inefficient conversion of profit from a narrow spread with efficient conversion. Finally, since $\alpha M = \tilde{M}$, the factor $h(\alpha M)$ in (4) becomes $h(\tilde{M})$ in (8), with h as in (5), and from here on a dot denotes the derivative with respect to the scaled time $\tilde{t}$ instead of t.

Table 2: The four dimensionless groups that remain after scaling and their meaning.

Symbol	Definition	Meaning
R	$\gamma s_0 \beta_0 I_0/(\mu\delta)$	Survival number: whether a first market maker in an empty market grows ($R > 1$) or shrinks ($R < 1$).
c	–	Liquidity response: how strongly the trading rate rises when the spread tightens.
b	$\beta_0/(\mu\alpha)$	How fast market makers use up trading interest, relative to how fast it expires.
ε	δ/μ	How fast market makers leave, relative to how fast trading interest expires. We assume $\varepsilon \ll 1$.

3 Equilibria and bifurcations

3.1 Equilibria and their stability

To determine the stability of the equilibria, we use the Jacobian of system (8),

$$J(\tilde{I},\tilde{M}) = \begin{pmatrix} -1 - b\tilde{M}(1+c\tilde{M}) & -b\tilde{I}(1+2c\tilde{M}) \\ \varepsilon R\, h(\tilde{M})\tilde{M} & \varepsilon\left(R\, h(\tilde{M})\tilde{I} - 1\right) + \dfrac{\varepsilon R(c-1)\tilde{I}\tilde{M}}{(1+\tilde{M})^2} \end{pmatrix}. \tag{9}$$

System (8) always has the empty-market equilibrium $(\tilde{I},\tilde{M}) = (1,0)$: there are no market makers, and the trading interest stays at the level I_0/μ where inflow and expiry balance. Since $h(0) = 1$, substituting $(1,0)$ into (9) gives

$$J(1,0) = \begin{pmatrix} -1 & -b \\ 0 & \varepsilon(R-1) \end{pmatrix}, \tag{10}$$

with eigenvalues -1 and $\varepsilon(R-1)$. The empty market is therefore stable for $R < 1$ and unstable for $R > 1$, as the interpretation of R as survival number predicts.

For positive equilibria, the $\tilde{M}$-equation in (8) is zero only if $R\, h(\tilde{M})\tilde{I} = 1$, since $\varepsilon\tilde{M} > 0$. This simplifies the second row of (9), and the Jacobian becomes

$$J = \begin{pmatrix} -1 - b\tilde{M}(1+c\tilde{M}) & -b\tilde{I}(1+2c\tilde{M}) \\ \varepsilon\tilde{M}/\tilde{I} & \varepsilon\tilde{M}\,\dfrac{c-1}{(1+\tilde{M})(1+c\tilde{M})} \end{pmatrix}. \tag{11}$$

To classify these equilibria, we first show that $\mathrm{tr}(J) < 0$, so that the sign of $\det(J)$ alone decides their stability. The upper-left entry of (11) is at most -1, because b, c and $\tilde{M}$ are non-negative. The lower-right entry is smaller than ε, because

$$\frac{\tilde{M}(c-1)}{(1+\tilde{M})(1+c\tilde{M})} < 1 \quad\Longleftrightarrow\quad (1+\tilde{M})(1+c\tilde{M}) - \tilde{M}(c-1) = 1 + 2\tilde{M} + c\tilde{M}^2 > 0,$$

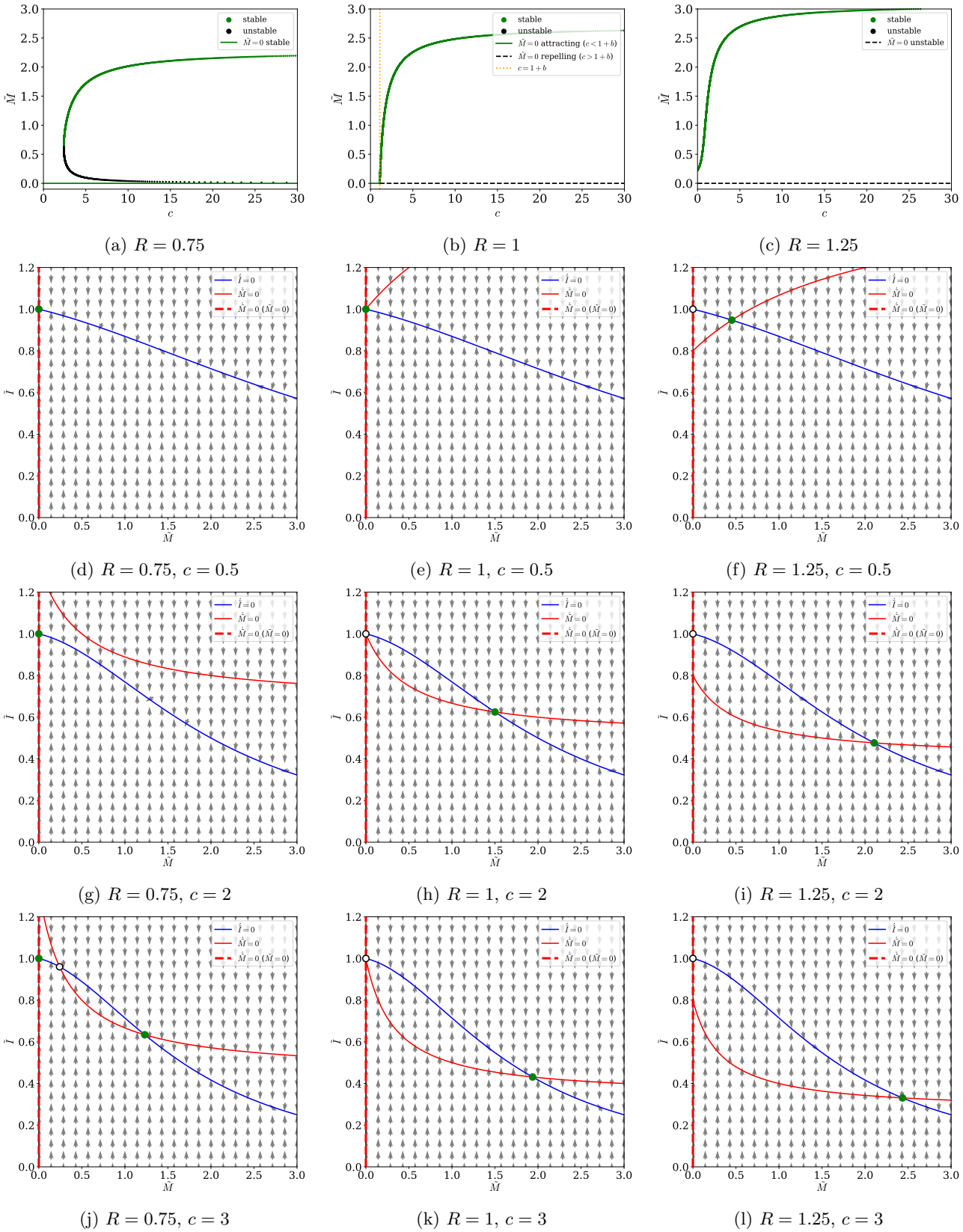
which holds for every $\tilde{M} \geq 0$, since the denominator on the left is positive. Together this gives $\mathrm{tr}(J) < -1+\varepsilon \leq 0$ for every $\varepsilon \leq 1$, and in particular for our small ε. Therefore a positive equilibrium is stable when $\det(J) > 0$ and a saddle when $\det(J) < 0$.

To find where the positive equilibria lie, we combine $R\, h(\tilde{M})\tilde{I} = 1$ with $\tilde{I} = 1/(1 + b\tilde{M}(1+c\tilde{M}))$ from the $\tilde{I}$-equation. Substituting and solving for R gives

$$R = \frac{(1+\tilde{M})\left(1 + b\tilde{M}(1+c\tilde{M})\right)}{1+c\tilde{M}} = \frac{bc\tilde{M}^3 + b(1+c)\tilde{M}^2 + (1+b)\tilde{M} + 1}{c\tilde{M}+1}. \tag{12}$$

The right-hand side is the value of R for which an industry of size $\tilde{M}$ is an equilibrium. For a given R this is a cubic equation in $\tilde{M}$, so there are at most three positive equilibria. We never find more than two numerically.

We now ask whether positive equilibria can also exist for $R < 1$, when the empty market is stable and a first market maker cannot survive on its own. Since (12) gives R as a function of $\tilde{M}$, this is the case exactly when its right-hand side drops below 1 for some $\tilde{M} > 0$. To see when this happens, we compare the numerator and the denominator of (12), which both contain the constant 1. For $c \leq 1 + b$ we have $(1+b)\tilde{M} \geq c\tilde{M}$, and the terms in $\tilde{M}^2$ and $\tilde{M}^3$ are positive, so the right-hand side is larger than 1 for every $\tilde{M} > 0$ and positive equilibria only exist for $R > 1$. For $c > 1 + b$ we have $(1+b)\tilde{M} < c\tilde{M}$, so the right-hand side drops below 1 when $\tilde{M}$ is small and positive equilibria also exist for $R < 1$.



(a) $R = 0.75$ (b) $R = 1$ (c) $R = 1.25$

(d) $R = 0.75$, $c = 0.5$ (e) $R = 1$, $c = 0.5$ (f) $R = 1.25$, $c = 0.5$

(g) $R = 0.75$, $c = 2$ (h) $R = 1$, $c = 2$ (i) $R = 1.25$, $c = 2$

(j) $R = 0.75$, $c = 3$ (k) $R = 1$, $c = 3$ (l) $R = 1.25$, $c = 3$

Figure 1: Top row: bifurcation diagrams of $\tilde{M}$ against c for three values of R; green dots are stable equilibria and black dots are unstable. The line at $\tilde{M} = 0$ is the empty market, solid green where it is stable or attracting and dashed black where it is unstable or repelling. In (b) the orange dotted line marks $c = 1 + b$. Other rows: phase portraits in the $(\tilde{M}, \tilde{I})$ plane for the indicated values of R and c, with the nullclines and the direction field. Green dots are stable equilibria and open dots are unstable. In all panels $b = 0.1$ and $\varepsilon = 0.01$. Since ε is small, the direction field is almost vertical.

3.2 Bifurcation diagrams and phase portraits

Equation (12) gives the equilibria only implicitly, so we now compute them numerically to see how they change with the two key parameters R and c. For the phase portraits, we write (12) as the cubic

$$bc\tilde{M}^3 + b(1+c)\tilde{M}^2 + (1+b-Rc)\tilde{M} + 1 - R = 0$$

and take its real positive roots $\tilde{M}^*$. The corresponding $\tilde{I}^*$ follows from the $\tilde{I}$-nullcline. We then evaluate the Jacobian (11) at $(\tilde{I}^*, \tilde{M}^*)$. Since $\mathrm{tr}(J) < 0$, the equilibrium is stable if $\det(J) > 0$ and a saddle if $\det(J) < 0$. The bifurcation diagrams show these equilibria and their stability as c varies.

We fix $b = 0.1$ and $\varepsilon = 0.01$. We do not fit the model to data, so these values only serve as an example. Since ε does not appear in (12), it does not change where the equilibria are, and since $\mathrm{tr}(J) < 0$ for every $\varepsilon \leq 1$, it does not change their stability either. It only changes how fast the system moves towards the equilibria.

Figure 1 shows the equilibria for three values of R. The top row contains bifurcation diagrams of $\tilde{M}$ against c, in which green dots are stable equilibria and black dots are unstable. The rows below show phase portraits for $c = 0.5$, 2 and 3.

For $R = 0.75$ (left column), the empty market is stable for every c. At $c \approx 2.4$ a pair of positive equilibria appears, one stable and one unstable. The phase portraits confirm this: for $c = 0.5$ and $c = 2$ every trajectory goes to the empty market, whereas for $c = 3$ there is also a stable positive equilibrium, which we call the liquid market, separated from the empty market by a saddle, which we call the threshold.

For $R = 1$ (middle column), the eigenvalue $\varepsilon(R-1)$ of $J(1,0)$ in (10) is zero. The linearization of (8) at the empty market is therefore inconclusive, so we look at the next, quadratic term in $\tilde{M}$. Close to $(1,0)$ we have $\tilde{I} \approx 1 - b\tilde{M}$ and $h(\tilde{M}) \approx 1 + (c-1)\tilde{M}$, so $\dot{\tilde{M}} \approx \varepsilon\,(c - (1+b))\tilde{M}^2$. For $c < 1 + b$ the empty market is therefore attracting, and by (12) it is the only equilibrium. For $c > 1 + b$ it is repelling, and Figure 1b shows a stable positive equilibrium that grows out of $\tilde{M} = 0$ at $c = 1 + b$.

For $R = 1.25$ (right column), the empty market is unstable and there is exactly one stable positive equilibrium, and therefore a liquid market, for every c.

3.3 Two-parameter diagram

Figure 2 combines these results in the (c, R) plane. For $c > 1 + b$, we write $R_\mathrm{f}(c)$ for the lowest value that the right-hand side of (12) takes over $\tilde{M} > 0$. This is the smallest R for which a positive equilibrium exists, and it is where the liquid market and the threshold appear together in a saddle-node bifurcation. We compute $R_\mathrm{f}(c)$ numerically as the minimum of the right-hand side of (12) over $\tilde{M} > 0$. The two lines divide the plane into three regions. In the green region ($R > 1$), the empty market is unstable and the liquid market is the only stable state. In the blue region, below $R_\mathrm{f}(c)$, only the empty market exists. In the orange region in between, both are stable, and the starting size of the industry decides which one is reached. The two lines meet at $(c, R) = (1 + b, 1)$, which is why $c > 1 + b$ is needed for this orange region to exist. A different b moves the meeting point $c = 1 + b$ and the curve $R_\mathrm{f}(c)$, but numerically we find the same three regions for every $b > 0$. For $R = 0.75$ the dashed line is crossed at $c \approx 2.4$, which agrees with Figure 1a. Since ε affects neither the position nor the stability of the equilibria, the figure holds for every $\varepsilon \leq 1$.

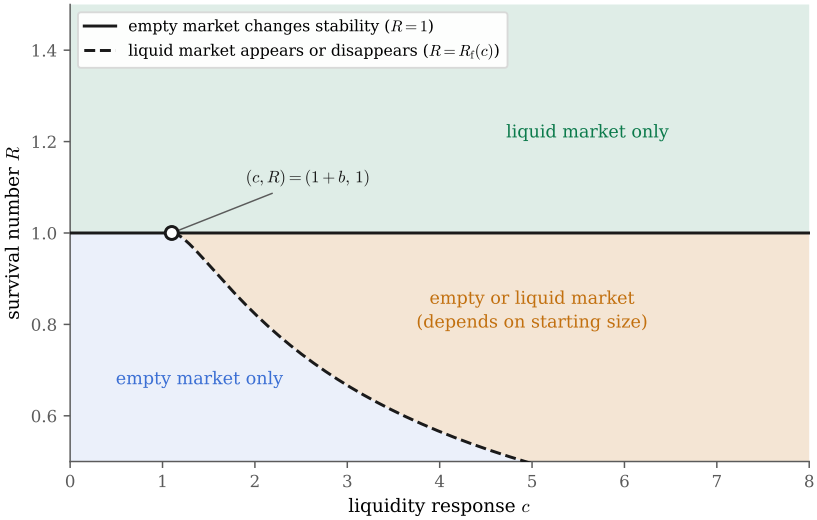


Figure 2: Equilibria in the (c, R) plane for $b = 0.1$. On the solid line $R = 1$ the empty market changes stability. On the dashed line $R = R_\mathrm{f}(c)$ the liquid market and the threshold appear or disappear together. The two lines meet at $(c, R) = (1 + b, 1)$. Green: only the liquid market is stable. Orange: both the empty and the liquid market are stable. Blue: only the empty market exists.

4 Interpretation of the results

4.1 Meaning of the equilibria

We now translate the analysis back to the market and answer our two questions. In our numerical results, the model has at most three equilibria, and each of them corresponds to a different state of the market. The first equilibrium is the empty market $(\tilde{I}, \tilde{M}) = (1, 0)$. In this case, there are no market makers, so the spread remains at its baseline s_0. Trading interest accumulates to I_0/μ and expires without being traded, making this an illiquid market. The second is the liquid market, the stable positive equilibrium. Here, the market-making presence is large and the spread $s_0/(1+\tilde{M})$ is tight. The scaled trading interest $\tilde{I}$ is below 1 (its value in the empty market), because orders are executed before they expire. For $R = 0.75$ and $c = 3$, Figure 1j shows this state at $\tilde{M} \approx 1.23$, so the spread is approximately 45 per cent of s_0. The third is the threshold, the unstable positive equilibrium, which lies at $\tilde{M} \approx 0.24$ in the same phase portrait. No market stays in this state, but it separates two outcomes: an industry that is smaller shrinks to zero, and an industry that is larger grows to the liquid market.

4.2 When can the industry sustain itself?

Our first question was when a market-making industry can sustain itself, and the simplest answer is when $R > 1$. By (7), in dimensional form this reads $\gamma s_0 \beta_0 I_0/\mu > \delta$, which means that the first market maker who enters an empty market and quotes the wide spread s_0 grows faster than it leaves. The empty market is then unstable and the liquid market is the only stable state (Figure 1c). The industry builds itself up from any starting size, and this holds for every value of c. This is the green region of Figure 2.

The case $R < 1$ is more interesting. A single entrant now loses money, so the industry cannot start from nothing. However, it can still exist if trading volume reacts strongly enough to a tighter spread: for $R = 0.75$ the liquid market appears at $c \approx 2.4$, while the empty market stays stable (Figure 1a). This is where the ecosystem engineering matters. A large industry keeps the spread tight, the tight spread brings in extra volume, and with that volume every market maker earns enough to stay. The industry keeps itself alive, but only if it is already there. Section 3 showed that this requires $c > 1 + b$, which says what the extra volume has to make up for: the 1 stands for the margin that is lost when the spread tightens, and $b = \beta_0/(\mu\alpha)$ for the trading interest that the market makers use up themselves. For $c \leq 1 + b$ a tighter spread does not pay off, and a market with $R < 1$ always ends up empty (the blue region of Figure 2).

4.3 Can the industry collapse abruptly?

Our second question was whether the industry can collapse abruptly. It can, but only for $c > 1 + b$. Suppose that R falls slowly, for example because the inflow of trading interest I_0 drops or because market makers leave faster. For $c < 1 + b$, the liquid market shrinks continuously to $\tilde{M} = 0$ as R decreases to 1, so the spread widens step by step and there is no sudden drop. For $c > 1 + b$, the industry stays large until R reaches the dashed line $R_\mathrm{f}(c)$ in Figure 2. There the liquid market and the threshold meet and disappear together, and the industry falls to zero. For $c = 2$ this happens at $R_\mathrm{f} \approx 0.8$, where still $\tilde{M} \approx 0.6$. Just before the collapse, the spread is about 63 per cent of s_0, so the market still looks healthy. Such a collapse is hard to undo: bringing R back above $R_\mathrm{f}(c)$ is not enough, because the empty market stays stable until $R > 1$. There is also a second way to collapse, in which the conditions do not change at all. When the empty and the liquid market are both stable, a sudden shock, such as many market makers leaving at once, can push the industry below the threshold. The liquid market still exists, but the industry is no longer large enough to reach it, and it shrinks to zero.

4.4 Link to ecosystem engineers and the flash crash

These results fit the picture of an ecosystem engineer from Section 1. A mussel bed has to be dense enough to slow the water down, and in the same way the industry has to be large enough to tighten the spread. The market makers reinforce themselves, because the liquidity they create improves the conditions for new market makers, who then add even more liquidity.

The model also gives a way to read the flash crash of 6 May 2010. Market makers withdrew within minutes, which is a sudden shock in $\tilde{M}$ of the second kind described above, but the market recovered afterwards (CFTC and SEC, 2010). In the model, a market recovers from such a shock in two cases: for $R > 1$, where the empty market is unstable, and for $R < 1$ in the orange region of Figure 2, as long as the shock does not push the industry below the threshold. The flash crash alone therefore cannot tell these two cases apart. This would be different if the shock was larger: for $R > 1$ the market makers always return, while in the orange region a shock that crosses the threshold empties the market for good. This risk is largest when R lies just above $R_\mathrm{f}(c)$, because the threshold is then close to the liquid market and even a small shock can push the industry below it. For instance, markets with a small inflow of trading interest I_0, or where market makers face high costs (small γ), have a lower R and therefore a higher risk of becoming illiquid after a sudden shock.

5 Discussion and conclusion

In this report we modelled market makers as ecosystem engineers that tighten the spread they earn from. The survival number R decides whether an empty market attracts market makers, and the liquidity response c decides whether a large industry can keep itself alive when $R < 1$. If $c > 1 + b$, an empty and a liquid market can both be stable, and the industry can collapse abruptly, either when conditions slowly get worse or after a sudden shock.

Our model has several limitations. Firstly, I and M are aggregate variables, while in reality market makers differ from each other, and so do the orders of traders, for example in size. Secondly, the ecosystem engineering in our model depends on the functions $\beta(s) = \beta_0(1 + c\alpha M)$ and $s(M) = s_0/(1 + \alpha M)$. These functions capture the idea that a larger market-making presence tightens the spread and increases trading volume, but they are not estimated from data. In particular, β grows without bound as M increases, while in real markets trading is likely to level off. A trading rate that levels off would make a large presence profit less from the extra volume and shift the positive equilibria. Thirdly, the parameter values are only examples, so we cannot say how large R or c is for a real market. In particular, the recovery after the flash crash of 2010 does not tell us whether the US equity markets lie in the green or the orange region of Figure 2. Finally, the model contains no randomness, while real shocks happen at random moments. It can therefore show that a shock may cause a lasting collapse, but not how likely such a shock is. Despite these limitations, the model captures how market makers create the liquidity they depend on, and when this can fail.

Future research could estimate R and c from data on spreads, trading volumes and the entry and exit of market makers, to find out which markets lie close to the region where a sudden collapse is possible. A second step would be a trading rate that levels off, or competition between different types of market makers, to test whether the threshold and the sudden collapse remain in a more realistic model. Adding random shocks would show how often a market in the orange region of Figure 2 would actually lose its market makers.

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AI usage statement

Claude Opus 5.5 (Anthropic) was used during this project. It helped us to write and debug the Python code for the phase portraits and bifurcation diagrams of Figure 1, as well as the (c, R) plane of Figure 2. We also used Claude for formatting in LaTeX as well as grammar and sentence clarity checks throughout. No AI tool was used for the model derivation, the mathematical analysis, or the interpretation of results. The code made by AI was tested extensively.

CRediT authorship statement

J.T. van Dijk: Writing – original draft (abstract and Section 5).
S.A.M. van Diermen: Formal analysis, Software, Visualization, Writing – original draft (Sections 1 and 4).
M.J. Esveldt: Formal analysis, Software, Visualization, Writing – original draft (Sections 2 and 3).
All authors: Conceptualization, Methodology, Writing – review & editing.